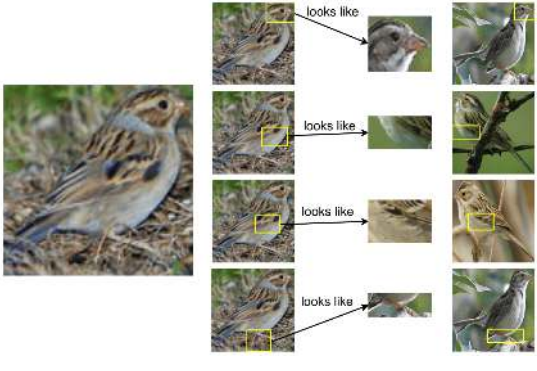
Beyond L_2: Generalizing Abductive Latent Explanations to Diverse Prototype-Based Architectures

Jules Soria, Alban Grastien, Romain Xu-Darme,
Julien Girard-Satabin, Zakaria Chihani, Daniela Cancila
CEA-List, Université Paris-Saclay

ECML PKDD

Prototypical-Parts Networks[1]

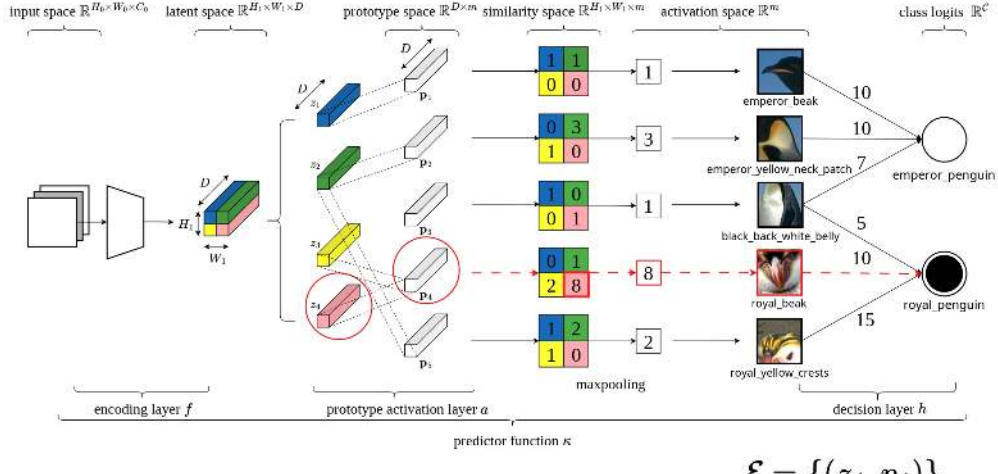


Sufficient Explanation ?

ProtoPNets give the k most activated prototypes as "evidence".
$\rightarrow k = 10$ is arbitrarily chosen.

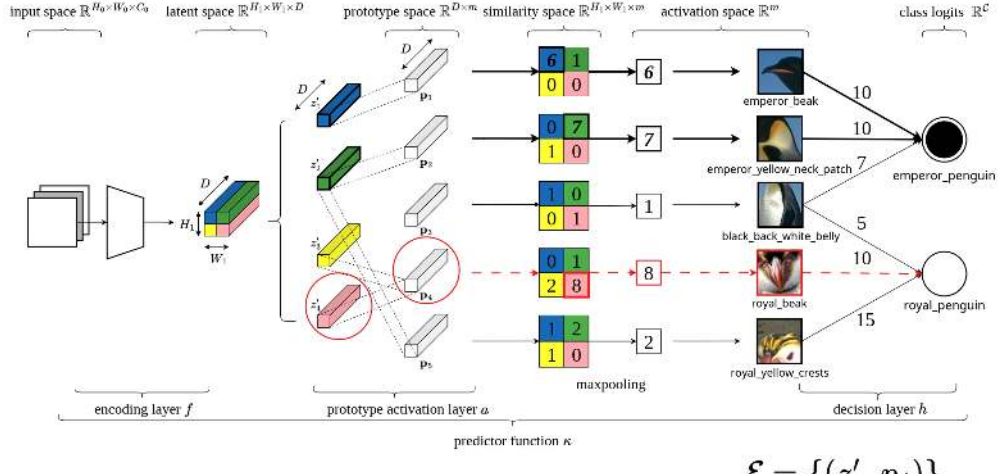
[1] Chaofan Chen et al. "This Looks Like That: Deep Learning for Interpretable Image Recognition". In: *NeurIPS*. 2019

Toy Example



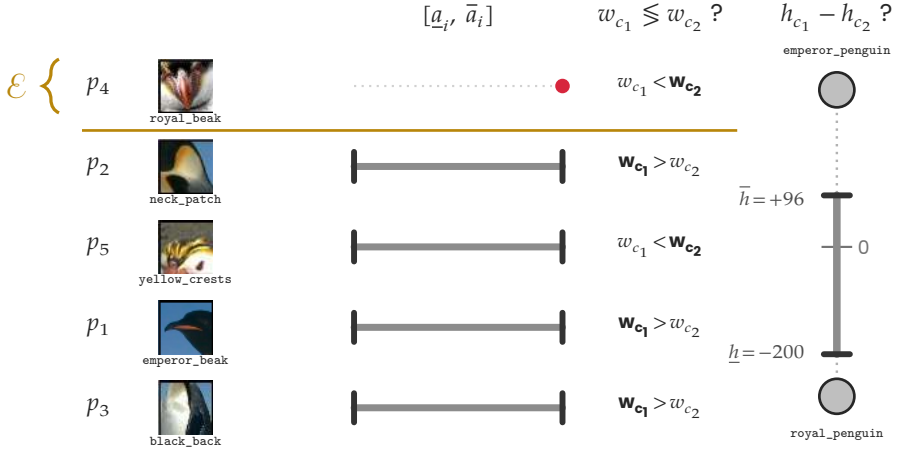
$$\mathcal{E} = \{(z_4, p_4)\}$$
$$\mathcal{E} \Rightarrow \kappa(\mathbf{x}) = 2?$$

Toy Counter-Example

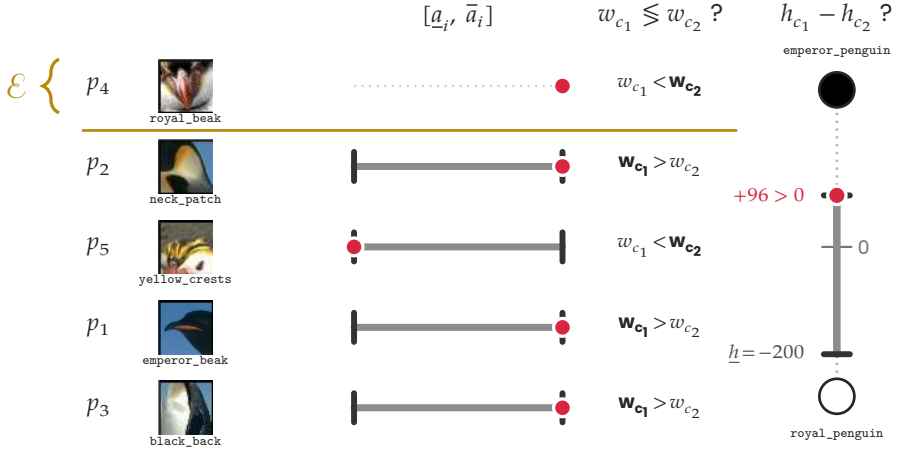


$$\mathcal{E} = \{(z'_4, p_4)\}$$
$$\mathcal{E} \not\Rightarrow \kappa(\mathbf{x}) = 2$$

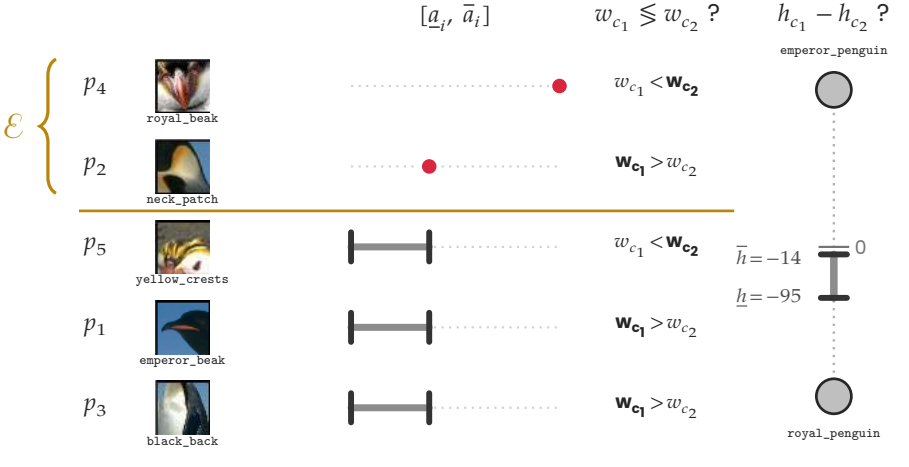
Why do we need bounds?



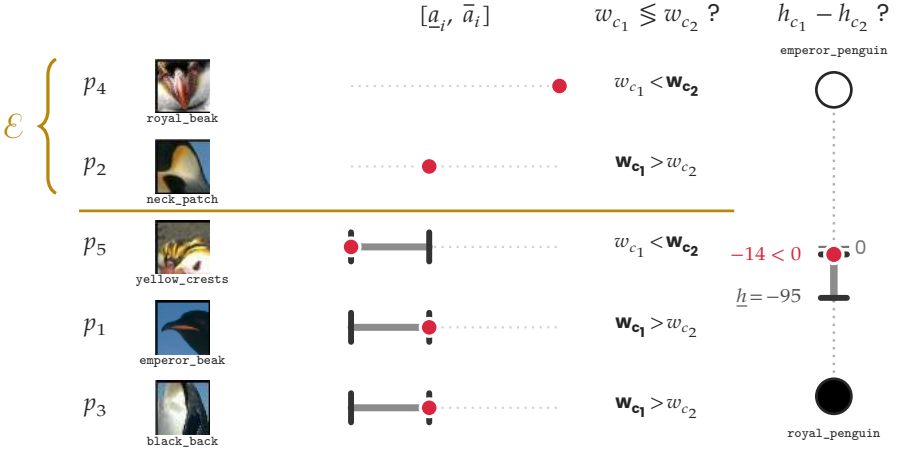
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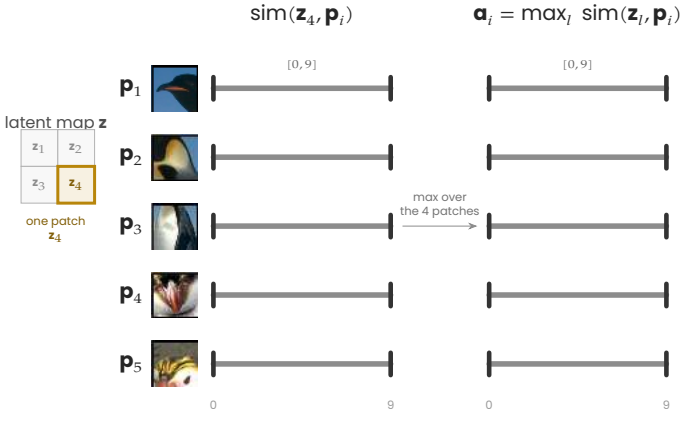
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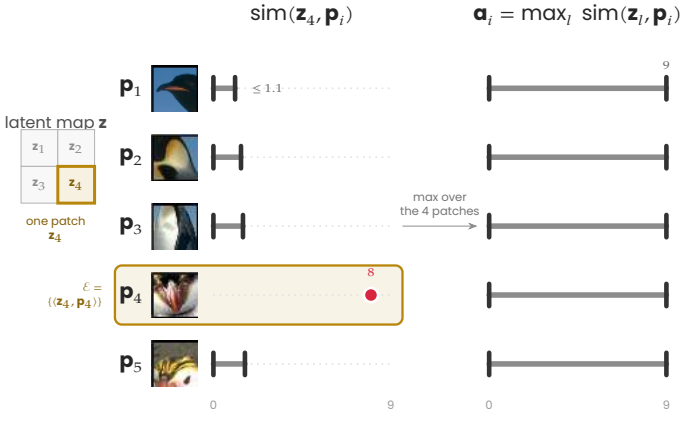
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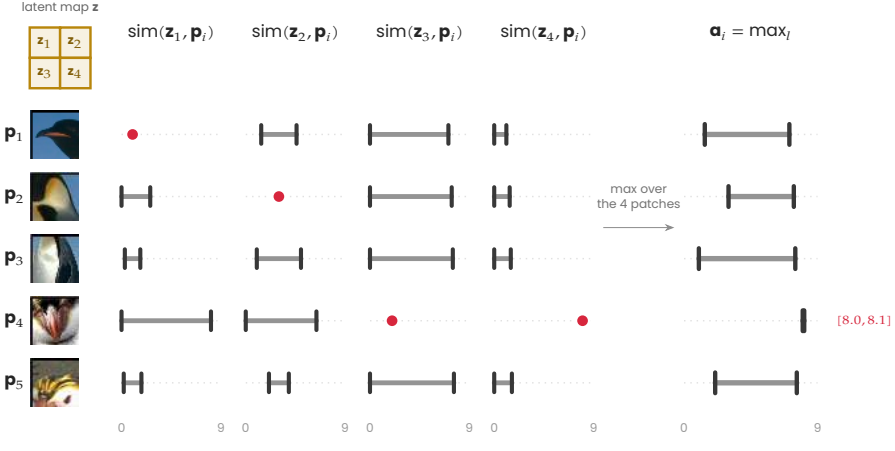
From activations to similarities



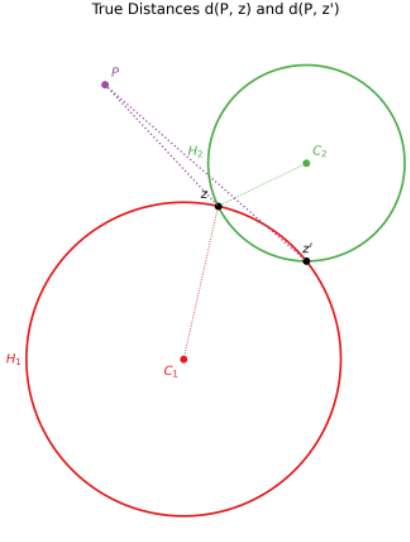
From activations to similarities

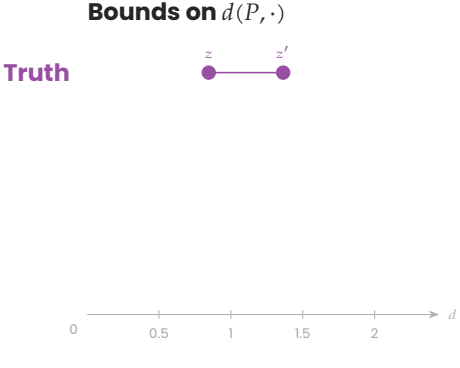


One pair per patch

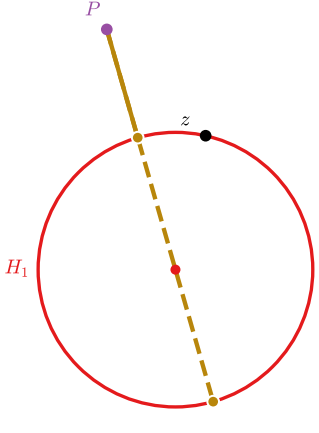


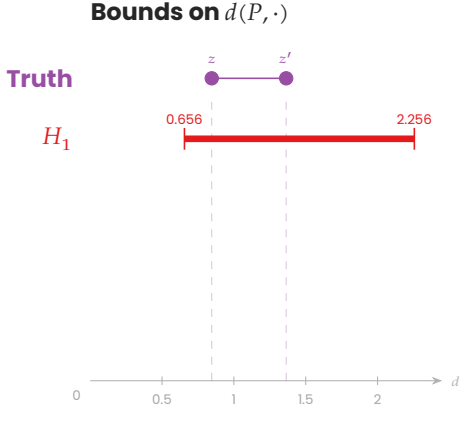
Bounding distances in the latent space



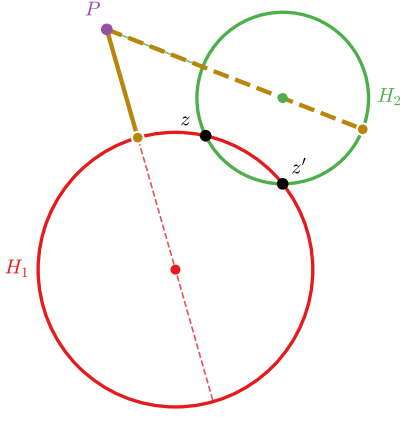


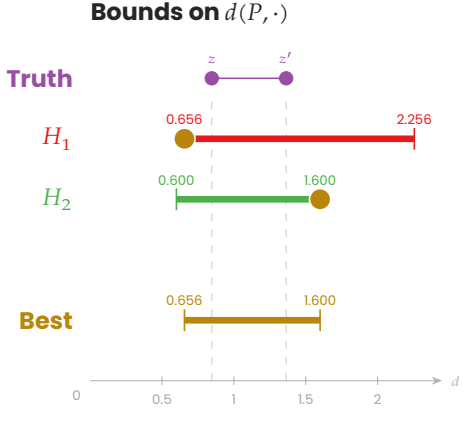
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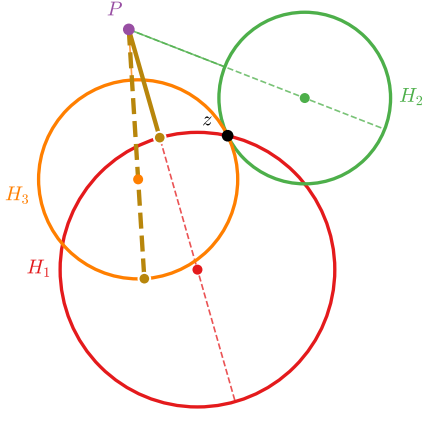


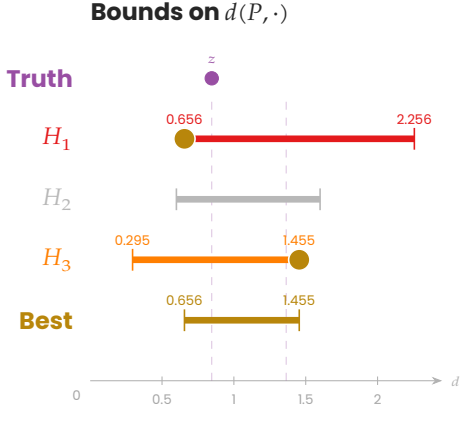
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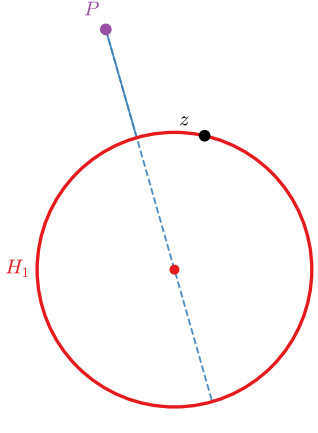


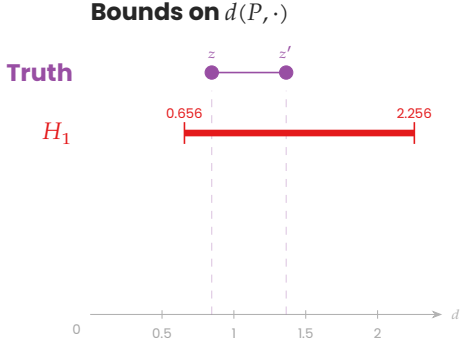
Bounding distances in the latent space



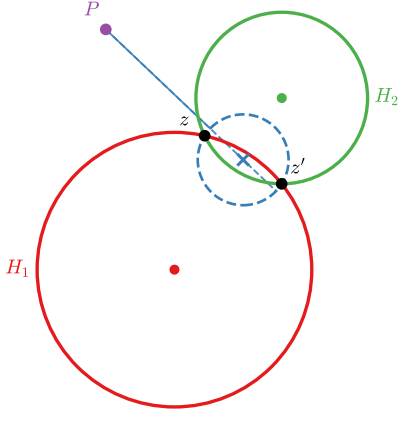


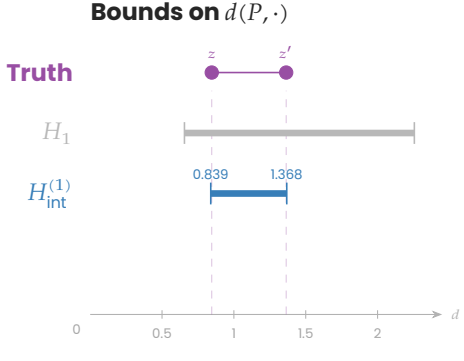
Bounding distances in the latent space



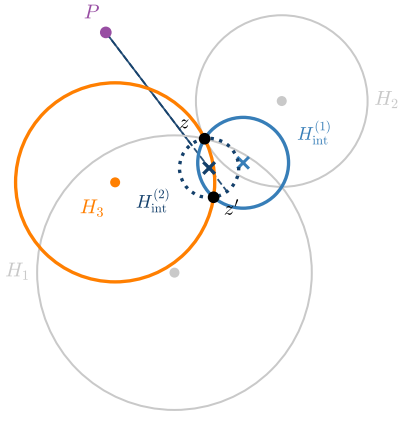


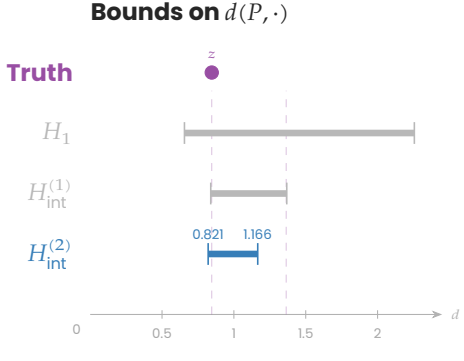
Bounding distances in the latent space





Bounding distances in the latent space





Modern PBNs are no longer Euclidean

Architecture	Similarity	Gaussian	Pooling
ProtoPNet	Euclidean	✗	Max
ProtoPool	Euclidean	✗	**Focal**
TesNet	**Cosine**	✗	Max
PIP-Net	**Cosine** *	✗	Max
MGProto	Euclidean	✓	Max
HyperPG	**Cosine**	✓	Max

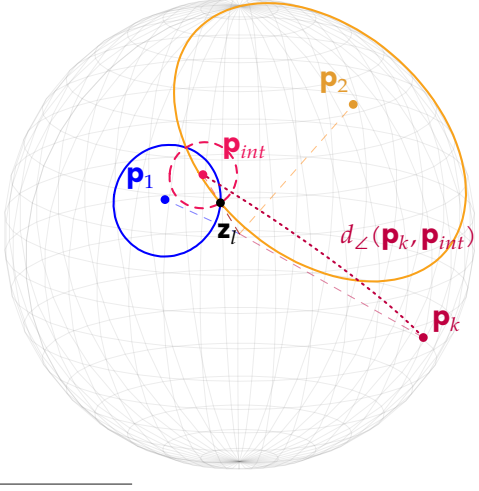
* dimensional projection: softmax over prototypes ⇒ simplex geometry

ALE[2] bounds assume a **Euclidean** latent space.

→ *map back* to existing bounds, or *construct new ones*.

[2] Jules Soria et al. "Formal Abductive Latent Explanations for Prototype-Based Networks". In: *Proceedings of the AAAI Conference on Artificial Intelligence*. Vol. 40. 30. 2026, pp. 25590–25598

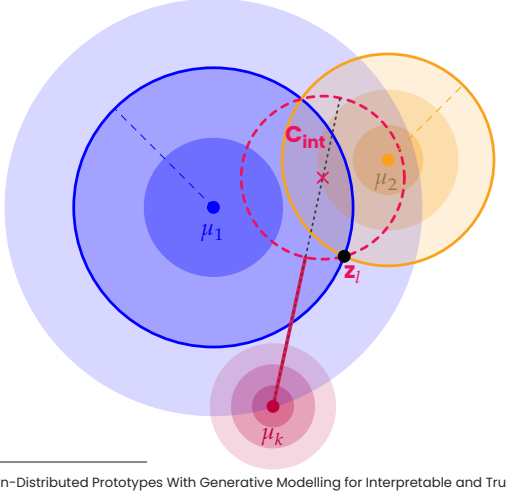
Generalizing Beyond L_2: Cosine Similarity[3,4]



[3] Jiaqi Wang et al. "Interpretable image recognition by constructing transparent embedding space". In: *CVPR*. 2021

[4] Frank Willard et al. "This Looks Better Than That: Better Interpretable Models with Protopnext". In: *arXiv:2406.14675* (2024)

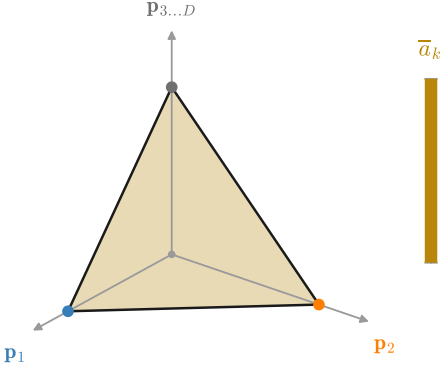
Generalizing Beyond L_2: Gaussian densities[5,6]



[5] Chong Wang et al. "Mixture of Gaussian-Distributed Prototypes With Generative Modelling for Interpretable and Trustworthy Image Recognition". In: *IEEE Transactions on Pattern Analysis and Machine Intelligence* 47.8 (2025), pp. 6974–6989. DOI: 10.1109/TPAMI.2025.3566425

[6] Nazanin Moradinasab et al. "Protogmm: Multi-prototype gaussian-mixture-based domain adaptation model for semantic segmentation". In: *2024 International Conference on Machine Learning and Applications (ICMLA)*. IEEE. 2024, pp. 99–106

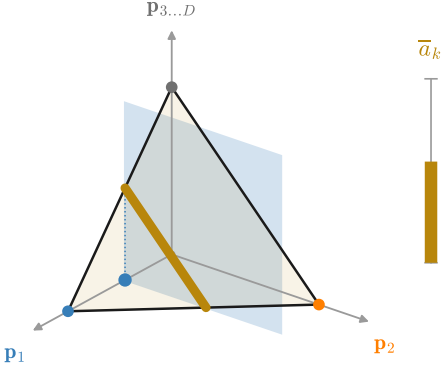
Generalizing Beyond L_2: Simplex (PIP-Net[7])



$$a_k \leq \max_l \left(1 - \sum_{\langle \mathbf{z}_l, \mathbf{p}_j \rangle \in \mathcal{E}} \text{sim}(\mathbf{z}_l, \mathbf{p}_j)\right)$$

[7] Meike Nauta et al. "Pip-net: Patch-based intuitive prototypes for interpretable image classification". In: *2023 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. IEEE. 2023, pp. 2744–2753

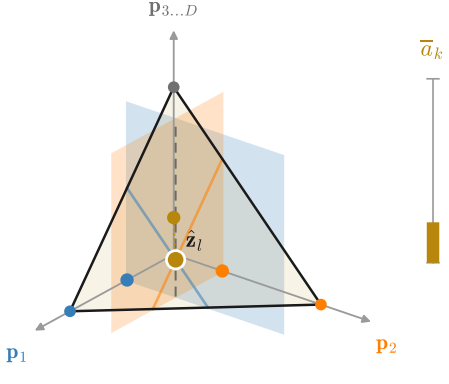
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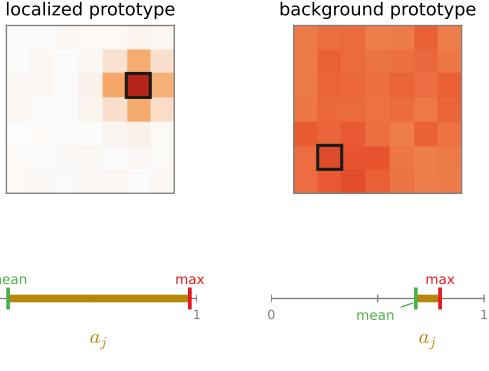


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[7] Meike Nauta et al. "Pip-net: Patch-based intuitive prototypes for interpretable image classification". In: *2023 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. IEEE. 2023, pp. 2744–2753

Generalizing Beyond L_2: Focal Pooling (ProtoPool[8])

Focal similarity changes only the *pooling*: $a_j = \max_{l \in \mathbf{L}} \text{sim}(\mathbf{z}_l, \mathbf{p}_j) - \mathop{\mathbb{E}}_{l \in \mathbf{L}} [\text{sim}(\mathbf{z}_l, \mathbf{p}_j)]$

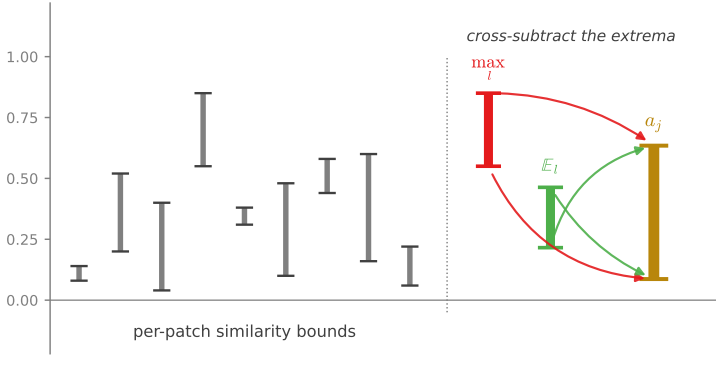


→ Works with *any* similarity function; no change to the geometric solver.

[8] Dawid Rymarczyk et al. "Interpretable Image Classification with Differentiable Prototypes Assignment". In: *ECCV*. 2022

Generalizing Beyond L_2: Focal Pooling (ProtoPool[8])

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[8] Dawid Rymarczyk et al. "Interpretable Image Classification with Differentiable Prototypes Assignment". In: *ECCV*. 2022

Experimental setup

Models & data

- **4 architectures**: ProtoPNet, PIP-Net, TesNet, isotropic-Gaussian ProtoPNet
- **3 datasets**: Oxford Flowers 102, Oxford-IIIT Pets, CUB-200
- All trained in `CaBRNet` [a]

[a] Romain Xu-Darme et al. "CaBRNet, an Open-Source Library for Developing and Evaluating Case-Based Reasoning Models". In: *Proc. xAI-2024 Workshops*. 2024. URL: https://cea.hal.science/cea-04688217

Generating an ALE

forward: add ⟨patch, prototype⟩ pairs until the prediction is locked
backward: drop every pair that turns out to be unnecessary
→ **subset-minimal**, no solver calls

Global explanations: which prototypes suffice?

	Flowers 102		Oxford-IIIT Pets	
Architecture	Top-k	Sparse	Top-k	Sparse
ProtoPNet	199 (19.5)	—	99 (26.6)	—
PIP-Net	**135** (17.6)	**103** (13.5)	131 (17.0)	**30** (4.0)
TesNet	295 (28.9)	—	**16** (4.4)	—
Gaussian	257 (25.2)	—	63 (17.1)	—

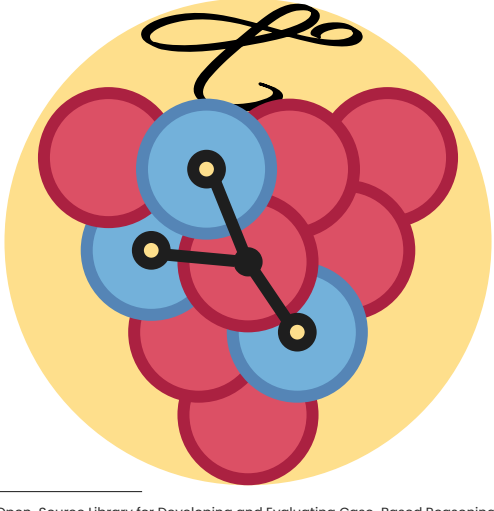
Absolute size (relative size, %); bold = best per column. All global ALEs return in under 2 s.
Grey: already available — Top-k is agnostic to the activation function. **Red**: enabled here.

Spatial explanations: which ⟨patch, prototype⟩ pairs?

	Flowers 102			Oxford-IIIT Pets		
Architecture	TI	HIA	Simplex	TI	HIA	Simplex
ProtoPNet	5273 (32.3)	675 (4.1)	—	7869 (43.4)	733 (4.0)	—
PIP-Net	—	—	250 (0.2)	—	—	190 (0.1)
TesNet	10438 (64.0)	**187** (1.1)	—	1750 (9.7)	**578** (3.2)	—
Gaussian	**1022** (6.3)	931 (5.7)	—	**1215** (6.7)	1126 (6.2)	—

Absolute size (relative size, %); bold = best per column.
Runtimes: TI 14–203 s, HIA 3–157 s, Simplex < 0.3 s.
TI: Triangle **I**nequality
HIA: Hypersphere **I**ntersection **A**pproximation
Grey: available for ProtoPNet only. **Red**: enabled here

CaBRNet : Case-Based Reasoning Networks[9]



[9] Romain Xu-Darme et al. "CaBRNet, an Open-Source Library for Developing and Evaluating Case-Based Reasoning Models". In: *Proc. xAI-2024 Workshops*. 2024. URL: https://cea.hal.science/cea-04688217

CaBRNet : Case-Based Reasoning Networks[9]

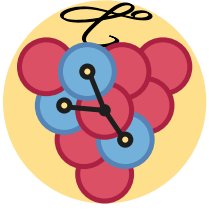
A Unified Library for Prototypical Networks

Instead of re-implementing baselines, CaBRNet provides a single, clean API for:

- ProtoPNet & PIP-Net
- ProtoTree &
- ProtoPool ...and more.

Quickstart:

```
uv pip install cabrnet
```



https://github.com/aiser-team/cabrnet

[9] Romain Xu-Darme et al. "CaBRNet, an Open-Source Library for Developing and Evaluating Case-Based Reasoning Models". In: *Proc. xAI-2024 Workshops*. 2024. URL: https://cea.hal.science/cea-04688217

Takeaway

One framework, four geometries

ALE was confined to **Euclidean** latent spaces. It now covers spherical (bounding caps), Gaussian (rescale → HIA → remap), simplex (residual mass) and focal pooling — all closed-form, solver-free, and **monotonically tightening**.

A directive for architects

First cross-architecture comparison of *formal* interpretability: **sparse, non-negative decision heads** are structurally friendlier to formal verification (PIP-Net wins on every dataset).

Next: train for provable interpretability, not just "by design".

jules.soria@cea.fr

Read the paper

References I

[1] Chaofan Chen et al. “This Looks Like That: Deep Learning for Interpretable Image Recognition”. In: *NeurIPS*. 2019.

[2] Jules Soria et al. “Formal Abductive Latent Explanations for Prototype-Based Networks”. In: *Proceedings of the AAAI Conference on Artificial Intelligence*. Vol. 40. 30. 2026, pp. 25590–25598.

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References II

[6] Nazanin Moradinasab et al. “Protogmm: Multi-prototype gaussian-mixture-based domain adaptation model for semantic segmentation”. In: *2024 International Conference on Machine Learning and Applications (ICMLA)*. IEEE. 2024, pp. 99–106.

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